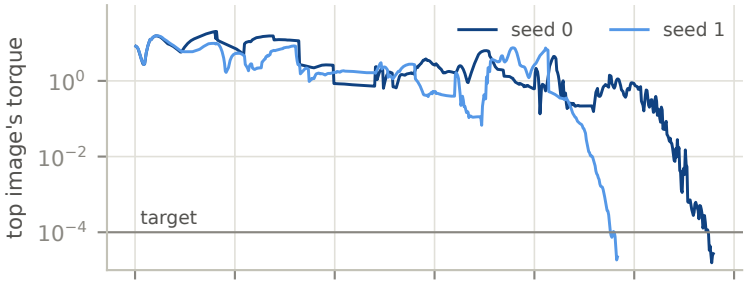
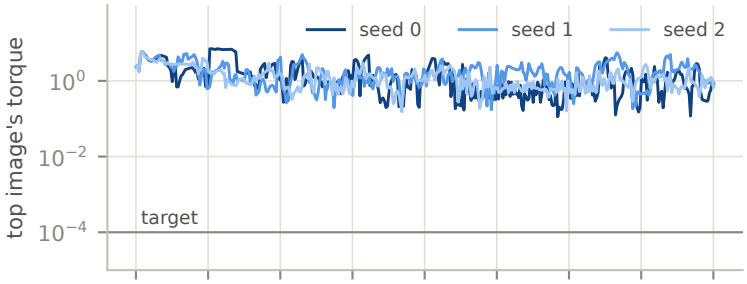
A Newton finish for GNEB: pilot

Newton–MINRES lands on the saddle exactly and cheaply, but this version does not clear the adoption bar. At 30³, all six finishes reach 22.478 k_BT with one negative mode in at most 53 field evaluations, after a chain of 3,600–7,700. Stopping the chain at a looser tolerance saves too little; stopping at the first lasting try would save 30–32%, but no run can know that moment in advance. At 15³ the landscape points to the cause of D49's failures: a staircase of passes and minima that 30³ does not show. The chain never settles there, and one try from its top image is unreliable. Each seed still reaches the highest pass on some tries (4–17 of 80). The mode count rejects minima and multi-nucleus saddles, and the pushes show which saddle joins the start.

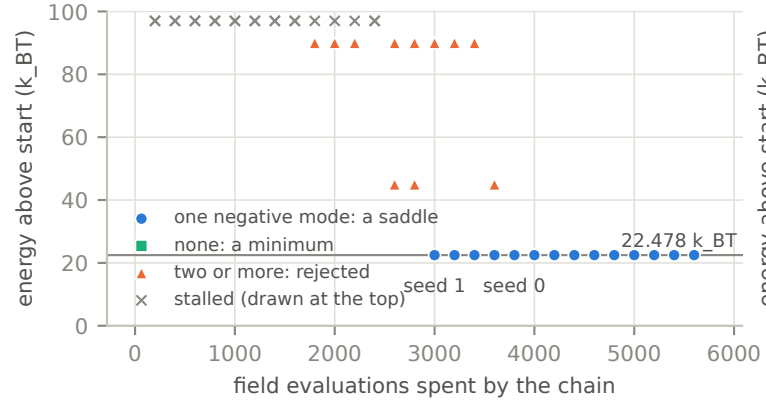
30³: the baseline chain converges



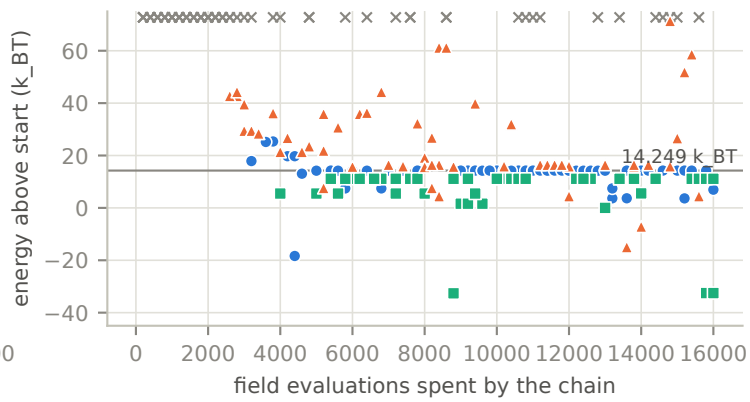
15³: the baseline chain never settles (2000 iterations)



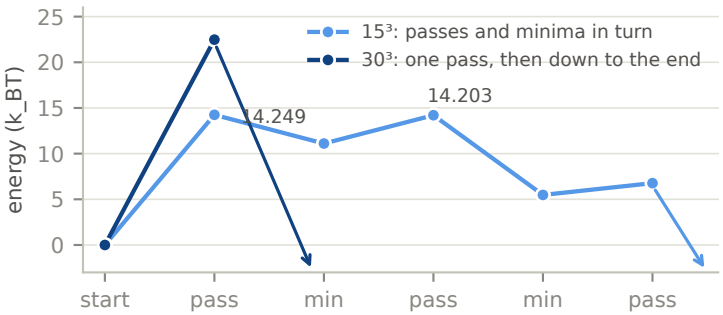
Stop the chain here and Newton lands at (labels: first lasting try)



Stop the chain here and Newton lands at



15³ is a staircase, 30³ a single pass



stationary points in order, each joined to the next by a push and relax

15³: Newton's tries along each chain, 80 per seed

seed	chain	from its top image	hit 14.249	any saddle
0	failed	stalled	17 of 80	29 of 80
1	failed	stalled	4 of 80	12 of 80
2	failed	minimum, 11.11	11 of 80	30 of 80

30³: total field evaluations, against the same seed's chain alone

seed	chain stops at	iterations	chain	Newton	modes	total	change	barrier	negative	pushes reach
0	1e-4 (baseline)	723	5798	—	—	5798	—	22.4783	—	—
0	0.01	638	5118	53	117	5288	-9%	22.4783	1	start, end
0	0.003	631	5062	25	117	5204	-10%	22.4783	1	start, end
0	0.001	965	7735	18	117	7870	+36%	22.4783	1	end, end
0	first lasting try	475	3800	147	117	4064	-30%	22.4783	1	not tried
1	1e-4 (baseline)	603	4838	—	—	4838	—	22.4783	—	—
1	0.01	448	3598	48	172	3818	-21%	22.4783	1	start, end
1	0.003	472	3790	18	172	3980	-18%	22.4783	1	start, end
1	0.001	473	3798	25	172	3995	-17%	22.4783	1	start, end
1	first lasting try	375	3000	139	172	3311	-32%	22.4783	1	not tried

The 40 nm $Nd_2Fe_{14}B$ cube of D32: $\mu_0 Ms$ = 1.61 T, K = 4.3 MJ/m³, A = 7.7 pJ/m, field −3.94 T 1° off z, 10 images, chain limit 2000 iterations, target torque 1e-4. MINRES to a relative residual of 1e-2, steps capped at 0.1 rad per cell, 50 Newton steps at most. Curvatures from Lanczos, the three lowest, so 3 means at least three. A field evaluation counts one, and so does a Hessian product. Tries are every 25 iterations of an unbroken baseline chain. The operator passes central differences (7e-11) and a macrospin's closed form. The landings at 44.96 and 89.99 k_BT are 2.00× and 4.00× the pass, which suggests nuclei at two and four corners; those states were not imaged. Not measured: the held-out wire and core-shell cube, any field but −3.94 T, and any stop rule a run could use without knowing the answer. Two seeds at 30³ is thin, and chain cost is erratic (seed 0 took more iterations at 1e-3 than at 1e-4). In one 30³ run of eight, both pushes relaxed to the end state at the same saddle: a 0.1 rad kick can overshoot. Decision D54. The method was dropped and its script removed; the operator and mode count live on in benchmarks/barrier_cell_size.py.